

Randomized Algorithms

The Probabilistic Method (I)

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Outline

- 1 Motivation
- 2 The Basic Counting Argument
- 3 The Expectation Argument
- 4 Derandomization Using Conditional Expectations
- 5 Sample and Modify
- 6 The Second Moment Method
- 7 The Conditional Expectation Inequality



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Motivation

- Prove the existence of objects.



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- Prove the existence of objects.
- If the probability of selecting an object with the required properties is **positive**, then the sample space must contain such an object.



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Example

- Coloring the edges of a graph with **two colors**.

Constraint: no large cliques with all edges having the same color.



Theorem 1

If $\binom{n}{k} 2^{-\binom{k}{2}+1} < 1$, then it is possible to color the edges of K_n with two colors so that it has no monochromatic K_k subgraph.



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- **Note:**

- There are $2^{\binom{n}{2}}$ possible colorings of K_n .
- There are $\binom{n}{k}$ different K_k cliques of K_n .

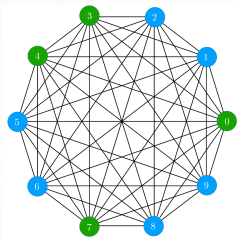


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- **Note:**

- There are $2^{\binom{n}{2}}$ possible colorings of K_n .
- There are $\binom{n}{k}$ different K_k cliques of K_n .
- Flip a fair coin independently to determine the color of each edge.



- $G_{\{v_0, v_3, v_4, v_7\}}$: a monochromatic K_4 .
- $G_{\{v_1, v_2, v_5, v_6, v_8, v_9\}}$: a monochromatic K_6 .
- No monochromatic K_k for $k \geq 7$.

General Idea

- Let A_i be the event that clique i is monochromatic.
- **Goal:** Prove that the probability is $\Pr \left[\bigcap_{i=1}^{\binom{n}{k}} \overline{A_i} \right] > 0$.



General Idea

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- **Goal:** Prove that the probability is $\Pr \left[\bigcap_{i=1}^{\binom{n}{k}} \overline{A_i} \right] > 0$.
- That is,

$$1 - \Pr \left[\bigcup_{i=1}^{\binom{n}{k}} A_i \right] > 0.$$



Proof

- **monochromatic:** Once the first edge in clique i is colored, the remaining $\binom{k}{2} - 1$ edges must all be given the same color. So,



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- By the union bound,

$$\Pr\left(\bigcup_{i=1}^{\binom{n}{k}} A_i\right) \leq \sum_{i=1}^{\binom{n}{k}} \Pr(A_i) = \binom{n}{k} 2^{-\binom{k}{2}+1} < 1,$$

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- The last inequality: the assumptions of the theorem.
- Hence

$$\Pr\left[\bigcap_{i=1}^{\binom{n}{k}} \overline{A_i}\right] = 1 - \Pr\left[\bigcup_{i=1}^{\binom{n}{k}} A_i\right] > 0.$$



For sufficiently large k

- Our calculations can be simplified if we note that:

For $n \leq 2^{k/2}$ and $k \geq 3$,

$$\begin{aligned}\binom{n}{k} 2^{-\binom{k}{2}+1} &\leq \frac{n^k}{k!} 2^{-(k(k-1)/2)+1} \\ &\leq \frac{2^{k/2+1}}{k!} \\ &< 1.\end{aligned}$$



A Monte Carlo construction

- Sample a coloring by coloring each edge independently at random.
 - How many samples must we generate before obtaining a sample satisfying our requirement?



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- **Example:** Finding a coloring on a graph of 1,000 vertices **with no monochromatic K_{20}** .
 - The probability that a random coloring has a monochromatic K_{20} is

$$\leq \frac{2^{20/2+1}}{20!} < 8.5 \cdot 10^{-16}.$$



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⇒ A Monte Carlo algorithm with a very small probability of failure.



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Question

What if we want a Las Vegas algorithm?

- Can we still have an expected polynomial running time randomized algorithm for generating such a sample?
 - A polynomial time verifier is required.
- Expected number of samples: $1/p$.
- For a fixed (constant) k , check all $\binom{n}{k}$ cliques and make sure they are not monochromatic.
 - Not polynomial time when k grows with n !
 - $O(n^k k^2) = O^*(n^{f(k)})$ time: **XP** (eXtendable Polynomial time).



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- To prove that an object with certain properties **exists**, we can design a probability space from which an element chosen at random yields an object with the desired properties with a **positive probability**.
- A similar and sometimes easier approach: Using an averaging argument.



The expectation (average) argument

- **Intuitive idea:** Say a discrete random variable X has $\mathbb{E}[X] = \mu$. Then there must be some value $a \leq \mu$ and $b \geq \mu$ such that $\Pr[X = a] > 0$ and $\Pr[X = b] > 0$.



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Lemma 1

Suppose we have a probability space S such and a random variable X defined on S such that $\mathbb{E}[X] = \mu$. Then

$$\Pr[X \geq \mu] > 0 \quad \text{and} \quad \Pr[X \leq \mu] > 0.$$



Proof: Let $\mathcal{X} = \{x \in \mathbb{R} : \Pr[X = x] > 0\}$ be the support of X .
Suppose that

$$\mu = \mathbb{E}[X] = \sum_{x \in \mathcal{X}} \Pr[X = x].$$

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If $\Pr[X \geq \mu] = 0$, then

$$\mu = \sum_{x \in \mathcal{X}} x \Pr[X = x] = \sum_{x < \mu} x \Pr[X = x] < \sum_{x < \mu} \mu \Pr[X = x] = \mu.$$

($\Rightarrow \Leftarrow$)



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Similarly, if $\Pr[X \leq \mu] = 0$ then

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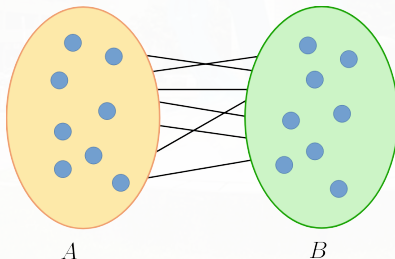
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Application 1: Finding a Large Cut

Theorem 2

Given an undirected graph $G = (V, E)$ with $|V| = n$ and $|E| = m$. There is a partition of V into disjoint sets A and B such that at least $m/2$ edges connect a vertex in A to a vertex in B . That is, there is a cut with value at least $m/2$.



Proof

- Construct sets A and B by randomly and independently assigning each vertex to one of the two sets.
- Let $e_1, e_2, \dots, e_m$ be an arbitrary enumeration of the edges of G .
- For $i = 1, \dots, m$, define X_i such that

$$X_i = \begin{cases} 1 & \text{if edge } i \text{ connects } A \text{ to } B \\ 0 & \text{otherwise} \end{cases}$$



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$$X_i = \begin{cases} 1 & \text{if edge } i \text{ connects } A \text{ to } B \\ 0 & \text{otherwise} \end{cases}$$

- The probability that edge e_i connects a vertex in A to a vertex in B is $\frac{1}{2}$, and thus

$$\mathbb{E}[X_i] = \frac{1}{2}$$



Proof (contd.)

- Let $C = (A, B)$ be a random variable denoting the value of the cut corresponding to the sets A and B . Then

$$\mathbb{E}[C(A, B)] = \mathbb{E}\left[\sum_{i=1}^m X_i\right] = \sum_{i=1}^m \mathbb{E}[X_i] = m \cdot \frac{1}{2} = \frac{m}{2}.$$



How to efficiently find a cut with value $\geq m/2$?

- We need “a lower bound” on the probability that a random partition has a cut of value at least $m/2$.
- To derive such a bound, let $p = \Pr [C(A, B) \geq \frac{m}{2}]$, and observe that $C(A, B) \leq m$, Then

$$\begin{aligned}\frac{m}{2} &= \mathbb{E}[C(A, B)] \\ &= \sum_{i < m/2} i \Pr[C(A, B) = i] + \sum_{i \geq m/2} i \Pr[C(A, B) = i] \\ &\leq (1 - p) \left(\frac{m}{2} - 1 \right) + pm,\end{aligned}$$



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which implies that $p \geq \frac{1}{m/2+1} \Rightarrow$ expected number of samples before finding a cut: $\leq m/2 + 1$.



Application 2: Maximum Satisfiability

The following expression is an instance of SAT:

$$(x_1 \vee \overline{x_2} \vee \overline{x_3}) \wedge (\overline{x_1} \vee \overline{x_3}) \wedge (x_1 \vee x_2 \vee x_4) \wedge (x_4 \vee \overline{x_3}) \wedge (x_4 \vee \overline{x_1}).$$



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Theorem 3

Given a set of m clauses, let k_i be the number of literals in the i th clause for $i = 1, 2, \dots, m$. Let $k = \min_{i \in \{1, 2, \dots, m\}} k_i$. Then there is a **truth assignment that satisfies at least**

$$\sum_{i=1}^m (1 - 2^{-k_i}) \geq m(1 - 2^{-k}) \quad \text{clauses.}$$



Proof

- Assign values independently and uniformly at random to the variables.
- The probability that the i th clause with k_i literals is satisfied is at least $(1 - 2^{-k})$.



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- The expected number of satisfied clauses is therefore at least

$$\sum_{i=1}^m (1 - 2^{-k_i}) \geq m(1 - 2^{-k}),$$

and there must be an assignment that satisfies at least that many clauses.



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 - x_i : the set where v_i is placed (so x_i is either A or B).
 - **Assumption:** The first k vertices have been placed. Consider the expected cut value if the *remaining* vertices are then placed *independently and uniformly into one of the two sets*.



Derandomization by Conditional Expectation (1/4)

- We show inductively how to place the next vertex so that

$$\mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_k] \leq \mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_{k+1}]$$



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- **The right-hand side:** the value of the cut determined by our placement algorithm.
- Hence our algorithm returns a cut whose value $\geq \mathbb{E}[C(A, B)] \geq m/2$.



Derandomization by Conditional Expectation (2/4)

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- Consider placing v_{k+1} randomly, so that it is placed in A or B with probability $1/2$ each.
- Let Y_{k+1} be a random variable representing the set where it is placed.



Derandomization by Conditional Expectation (3/4)

- Then

$$\begin{aligned}\mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_k] &= \frac{1}{2}\mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_k, Y_{k+1} = A] \\ &+ \frac{1}{2}\mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_k, Y_{k+1} = B].\end{aligned}$$



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- It follows that

$$\begin{aligned}\max \left(\mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_k, Y_{k+1} = A], \mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_k, Y_{k+1} = B] \right) \\ \geq \mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_k].\end{aligned}$$



Derandomization by Conditional Expectation (4/4)

- Compute the two quantities

- $\mathbb{E}[C(A, B) \mid x_1, \dots, x_k, Y_{k+1} = A]$
- $\mathbb{E}[C(A, B) \mid x_1, \dots, x_k, Y_{k+1} = B]$

and then place v_{k+1} in the set that yields the larger expectation.

- Once we do this, we will have a placement satisfying

$$\mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_k] \leq \mathbb{E}[C(A, B) \mid x_1, x_2, \dots, x_{k+1}].$$

Computation of $\mathbb{E}[C(A, B) \mid x_1, \dots, x_k, Y_{k+1} = A \text{ (or } B)]$

- For the first $k + 1$ vertices, compute the number of edges that contribute to the cut.
- For all the other edges, each one contributes to the cut with prob. $\frac{1}{2}$.



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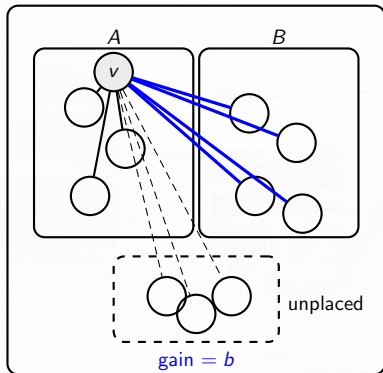
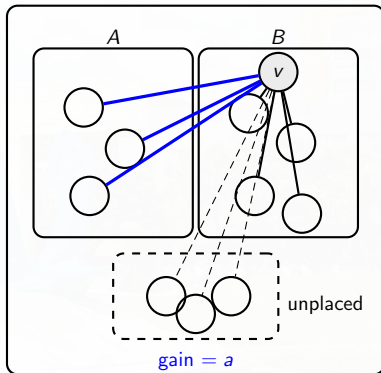
- For the first $k + 1$ vertices, compute the number of edges that contribute to the cut.
- For all the other edges, each one contributes to the cut with prob. $\frac{1}{2}$.
 - Depends on whether v_{k+1} has more neighbors in A or in B .



Simple greedy derandomized algorithm

- ① Take the vertices in some order.
- ② Place the first vertex arbitrarily in A .
- ③ Place each successive vertex to maximize the number of edges crossing the cut.
 - Equivalently, place each vertex on the side with fewer neighbors, breaking ties arbitrarily.



Two choices for v_{k+1} Place v_{k+1} in A Place v_{k+1} in B 

$$\mathbb{E}[C \mid x_1, x_2, \dots, x_k, v_{k+1} = A] = b, \quad \mathbb{E}[C \mid x_1, x_2, \dots, x_k, v_{k+1} = B] = a.$$

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Sample and Modify

Sample and Modify (Two Stages)

- 1st Stage: Construct a random structure that does not have the required properties.
- 2nd Stage: Modify the structure so that the required properties are satisfied.



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Sample and Modify (Two Stages)

- 1st Stage: Construct a random structure that does not have the required properties.
- 2nd Stage: Modify the structure so that the required properties are satisfied.
- In some cases, it is easy to work using this indirect approach.



Application 1: Independent Sets

Theorem 4

Let $G = (V, E)$ be a graph on n vertices with m edges. Then G has an independent set with at least $n^2/4m$ vertices.

Proof: Let $d = 2m/n$ be the average degree of the vertices in G . Consider the following randomized algorithm.



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Proof: Let $d = 2m/n$ be the average degree of the vertices in G . Consider the following randomized algorithm.

- 1 Delete each vertex of G (together with its incident edges) independently with probability $1 - 1/d$.
- 2 For each remaining edge, remove it and one of its adjacent vertices.



Application 1: Independent Sets

Theorem 4

Let $G = (V, E)$ be a graph on n vertices with m edges. Then G has an independent set with at least $n^2/4m$ vertices.

Proof: Let $d = 2m/n$ be the average degree of the vertices in G . Consider the following randomized algorithm.

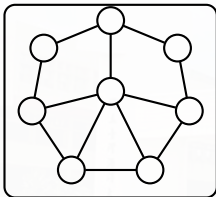
- 1 Delete each vertex of G (together with its incident edges) independently with probability $1 - 1/d$.
- 2 For each remaining edge, remove it and one of its adjacent vertices.

The remaining vertices form an independent set (all edges have been removed).



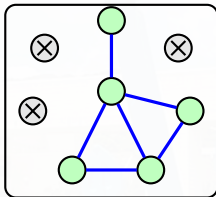
Sample & Modify for Independent Sets

1. Original graph G 2. After random sampling 3. After modification



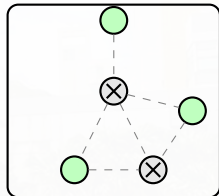
Delete each vertex independently

with survival probability $1/d$.



Let X be the number of surviving vertices,

and let Y be the number of surviving edges.



Remove one endpoint from each surviving edge.

The remaining green vertices form an independent set.

$$\mathbb{E}[X] = \frac{n}{d}, \quad \mathbb{E}[Y] = \frac{n}{2d}, \quad \Rightarrow \quad \mathbb{E}[X - Y] = \frac{n}{2d} = \frac{n^2}{4m}.$$



Proof (2/3)

- Let X be the number of vertices that survive the first step of the algorithm.
- Since the graph has n vertices and each vertex survives with probability $1/d$, it follows that

$$\mathbb{E}[X] = \frac{n}{d}.$$



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- Let Y be the number of edges that survive the first step. There are $nd/2$ edges in the graph, and an edge **survives** iff its **two adjacent vertices survive**.
- Thus,

$$\mathbb{E}[Y] = \frac{nd}{2} \left(\frac{1}{d}\right)^2 = \frac{n}{2d}.$$



Proof (3/3)

- The second step of the algorithm removes all the remaining edges and at most Y vertices.
- When the algorithm terminates, it outputs an independent set of size at least $X - Y$, and

$$\mathbb{E}[X - Y] = \frac{n}{d} - \frac{n}{2d} = \frac{n}{2d}.$$

- The expected size of the independent set generated by the algorithm is $n/2d$, so the graph has an independent set with at least $n/2d = n^2/4m$ vertices.



Application 2: Graphs with Large Girth

Theorem 5

For any integer $k \geq 3$, there is a graph with n nodes, at least $\frac{1}{4}n^{1+1/k}$ edges, and girth at least k .

- **girth**: the length of a **shortest cycle** contained in the graph.



Application 2: Graphs with Large Girth

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- **girth**: the length of a **shortest cycle** contained in the graph.

Proof: We first sample a random graph $G \in G_{n,p}$ with $p = n^{1/k-1}$. Let X be the number of edges in the graph. Then

$$\mathbb{E}[X] = p \binom{n}{2} = \frac{1}{2} \left(1 - \frac{1}{n}\right) n^{1/k+1}.$$

- $G_{n,p}$: a set of graphs of n vertices where every (u, v) exists with prob. p .



Proof (2/3)

- Let Y be the number of cycles in the graph of length $\leq k - 1$.
- Any specific possible cycle of length i , where $3 \leq i \leq k - 1$, occurs with probability p^i .



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- Also, there are $\binom{n}{i} \frac{(i-1)!}{2}$ possible cycles of length i .
 - First, consider choosing the i vertices, then consider the possible orders, and finally keep in mind that **reversing the order yields the same cycle**.



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- Also, there are $\binom{n}{i} \frac{(i-1)!}{2}$ possible cycles of length i .
 - First, consider choosing the i vertices, then consider the possible orders, and finally keep in mind that **reversing the order yields the same cycle**.
- Hence, (recall that $p = n^{1/k-1}$)

$$\begin{aligned}
 \mathbb{E}[Y] &= \sum_{i=3}^{k-1} \binom{n}{i} \frac{(i-1)!}{2} p^i \leq \sum_{i=3}^{k-1} n^i p^i = \sum_{i=3}^{k-1} n^i (n^{1/k-1})^i \\
 &= \sum_{i=3}^{k-1} n^{i/k} < kn^{(k-1)/k}.
 \end{aligned}$$



Proof (3/3)

X : # edges in the graph; Y : # cycles in the graph of length $\leq k - 1$.

- We modify the original randomly chosen graph G by **eliminating one edge from each cycle of length $\leq k - 1$** .
- The modified graph has girth at least k ($\because$ **shorter cycles are destroyed**).



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- We modify the original randomly chosen graph G by **eliminating one edge from each cycle of length $\leq k - 1$** .
- The modified graph has girth at least k ($\because$ **shorter cycles are destroyed**).
- When n is sufficiently large, the expected number of edges in the resulting graph is

$$\mathbb{E}[X - Y] \geq \frac{1}{2} \left(1 - \frac{1}{n}\right) n^{1/k+1} - kn^{(k-1)/k} \geq \frac{1}{4} n^{1/k+1}.$$

- So, there exists a graph with $\geq \frac{1}{4} n^{1+1/k}$ edges and girth $\geq k$.



Outline

- 1 Motivation
- 2 The Basic Counting Argument
- 3 The Expectation Argument
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- 6 The Second Moment Method**
- 7 The Conditional Expectation Inequality



The Second Moment Method

- In the $G_{n,p}$ model, it is often the case that there is a threshold function f such that
 - when $p = O(f(n))$ or $p = o(f(n))$, **almost no** graph has the desired property;
 - when $p = \Omega(f(n))$ or $p = \omega(f(n))$, **almost every** graph has the desired property.



The Second Moment Method

- The following theorem from Chebyshev's inequality is often used.

Theorem 6 [Derived from Chebyshev's Inequality]

X is a nonnegative integer-valued random variable, then

$$\Pr[X = 0] \leq \frac{\text{Var}[X]}{(\mathbb{E}[X])^2}.$$



The Second Moment Method

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Theorem 6 [Derived from Chebyshev's Inequality]

X is a nonnegative integer-valued random variable, then

$$\Pr[X = 0] \leq \frac{\text{Var}[X]}{(\mathbb{E}[X])^2}.$$

Proof:

$$\Pr[X = 0] \leq \Pr[|X - \mathbb{E}[X]| \geq \mathbb{E}[X]] \leq \frac{\text{Var}[X]}{(\mathbb{E}[X])^2}.$$



Application: Threshold Behavior in Random Graphs

Theorem 7

Consider $G_{n,p}$, suppose that $p = f(n)$.

Let A be the event that a random graph chosen from $G_{n,p}$ has a clique of ≥ 4 vertices. Then, for any $\varepsilon > 0$ and sufficiently large n ,

$$\Pr[A] < \varepsilon \quad \text{if } f(n) = o(n^{-2/3}).$$

Similarly, if $f(n) = \omega(n^{-2/3})$ then, for sufficiently large n ,

$$\Pr[\bar{A}] < \varepsilon.$$



Proof (1/4)

- We first consider the case in which $p = f(n)$ and $f(n) = o(n^{-2/3})$.
- $C_1, C_2, \dots, C_{\binom{n}{4}}$: an enumeration of all the subsets of 4 vertices in G .
- Let

$$X_i = \begin{cases} 1 & \text{if } C_i \text{ is a 4-clique} \\ 0 & \text{otherwise.} \end{cases}$$

- Let

$$X = \sum_{i=1}^{\binom{n}{4}} X_i,$$

so that (A 4-clique has 6 edges)

$$\mathbb{E}[X] = \binom{n}{4} p^6$$



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so that (A 4-clique has 6 edges)

$$\mathbb{E}[X] = \binom{n}{4} p^6 = o(1).$$



Proof (2/4)

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- So, in this case $\mathbb{E}[X] = o(1) \Rightarrow \mathbb{E}[X] < \varepsilon$ for sufficiently large n .
- Since X is a nonnegative integer-valued random variable, it follows that $\Pr[X \geq 1] \leq \mathbb{E}[X] < \varepsilon$.



Proof (2/4)

- So, in this case $\mathbb{E}[X] = o(1) \Rightarrow \mathbb{E}[X] < \varepsilon$ for sufficiently large n .
- Since X is a nonnegative integer-valued random variable, it follows that $\Pr[X \geq 1] \leq \mathbb{E}[X] < \varepsilon$.
- Hence, the probability that a random graph chosen from $G_{n,p}$ has a clique of ≥ 4 vertices is less than ε .



Proof (3/4)

- Next, consider the case when $p = f(n) = \omega(n^{-2/3})$.
- $\mathbb{E}[X] \rightarrow \infty$ as n grows large.
 - Not sufficient to conclude that, with high probability, a graph chosen random from $G_{n,p}$ has a clique of ≥ 4 vertices.



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- Then, we must show that $\text{Var}[X] = o((\mathbb{E}[X])^2)$.
- Consider the following lemma:



Proof (4/4)

Lemma 2

Let $Y_i, i = 1, 2, \dots, m$, be 0-1 random variables, and let $Y = \sum_{i=1}^m Y_i$.
Then

$$\text{Var}[Y] \leq \mathbb{E}[Y] + \sum_{1 \leq i < j \leq m; i \neq j} \text{Cov}(Y_i, Y_j).$$



Proof of Lemma 2

- For any sequence of random variables $Y_1, Y_2, \dots, Y_m$,

$$\text{Var} \left[\sum_{i=1}^m Y_i \right] = \sum_{i=1}^m \text{Var}[Y_i] + \sum_{1 \leq i, j \leq m; i \neq j} \text{Cov}(Y_i, Y_j).$$

- This is the generalization from two variables to m variables.
- When Y_i is a 0-1 random variable, $\mathbb{E}[Y_i^2] = \mathbb{E}[Y_i]$ and so

$$\text{Var}[Y_i] = \mathbb{E}[Y_i^2] - (\mathbb{E}[Y_i])^2 \leq \mathbb{E}[Y_i],$$

giving the lemma.



Finishing the proof of Theorem 7

Theorem 7

Consider $G_{n,p}$, suppose that $p = f(n)$.

Let A be the event that a random graph chosen from $G_{n,p}$ has a clique of ≥ 4 vertices. Then, for any $\varepsilon > 0$ and sufficiently large n ,

$$\Pr[A] < \varepsilon \quad \text{if } f(n) = o(n^{-2/3}).$$

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- Compute variance of the number of 4-cliques

$$\text{Var}[X] = \text{Var} \left(\sum_{i=1}^{\binom{n}{4}} X_i \right).$$



Using covariance

- Since X_i is an indicator,

$$\text{Var}[X_i] \leq \mathbb{E}[X_i] = p^6, \quad \Rightarrow \quad \sum_i \text{Var}(X_i) \leq \binom{n}{4} p^6.$$

- The remaining term is the total covariance. It depends on how much the two 4-cliques C_i and C_j overlap.
- Write $C_i \cap C_j$ for their intersection. We consider the cases $|C_i \cap C_j| = 0, 1, 2, 3$.



Pairs of cliques: overlap cases (1/2)

- **Case** $|C_i \cap C_j| = 0$ or 1.

- The sets of edges involved in the two cliques are disjoint.
- Thus X_i and X_j are independent, so $\mathbb{E}[X_i X_j] - \mathbb{E}[X_i] \mathbb{E}[X_j] = 0$.

- **Case** $|C_i \cap C_j| = 2$.

- The cliques share one edge; altogether $\binom{4}{2} - 1 = 11$ distinct edges must be present.
- Hence $\mathbb{E}[X_i X_j] \leq p^{11} \Rightarrow \mathbb{E}[X_i X_j] - \mathbb{E}[X_i] \mathbb{E}[X_j] \leq p^{11}$.
- The number of ordered pairs (C_i, C_j) with $|C_i \cap C_j| = 2$ is

$$\binom{n}{6} \binom{6}{2, 2, 2}.$$

Choose 6 vertices and then split them into $C_i \cap C_j$ (2 vertices), 2 vertices for $C_i \setminus C_j$, and 2 for $C_j \setminus C_i$.



Pairs of cliques: overlap cases (2/2)

- **Case** $|C_i \cap C_j| = 3$.

- The cliques share three vertices (a triangle), hence $\binom{4}{2} - 3 = 9$ distinct edges must appear.
- Thus

$$\mathbb{E}[X_i X_j] - \mathbb{E}[X_i] \mathbb{E}[X_j] \leq \mathbb{E}[X_i X_j] \leq p^9.$$

- There are

$$\binom{n}{5} \binom{5}{3, 1, 1}$$

ordered pairs (C_i, C_j) with $|C_i \cap C_j| = 3$.



Completing the bound on $\text{Var}(X)$

- Collecting all contributions,

$$\text{Var}(X) \leq \binom{n}{4} p^6 + \binom{n}{6} \binom{6}{2,2,2} p^{11} + \binom{n}{5} \binom{5}{3,1,1} p^9.$$

- Using $p = f(n) = \omega(n^{-2/3})$ and $\mathbb{E}[X] = \binom{n}{4} p^6$, one checks that

$$\text{Var}(X) = o(n^8 p^{12}) = o((\mathbb{E}[X])^2).$$

since $(\mathbb{E}[X])^2 = (\binom{n}{4} p^6)^2 = \Theta(n^8 p^{12})$.



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By the second moment method, this implies $\Pr[X = 0] = o(1)$,
 $\Rightarrow$ with high probability G contains a copy of K_4 .



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- For a **sum of Bernoulli random variables**, there is an alternative to the second moment method.

Theorem 8

Let $X = \sum_{i=1}^n X_i$, where each X_i is a 0-1 random variable. Then

$$\Pr[X > 0] \geq \sum_{i=1}^n \frac{\Pr[X_i = 1]}{\mathbb{E}[X \mid X_i = 1]}.$$

- Note that X_i 's need not to be independent here.



Proof

- Let $Y = 1/X$ if $X > 0$, with $Y = 0$ otherwise. Then

$$\mathbb{E}[XY]$$



Proof

- Let $Y = 1/X$ if $X > 0$, with $Y = 0$ otherwise. Then

$$\mathbb{E}[XY] = \sum_{x>0} x \cdot \frac{1}{x} \cdot \Pr[X = x] = \sum_{x>0} \Pr[X = x]$$



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$$\mathbb{E}[XY] = \sum_{x>0} x \cdot \frac{1}{x} \cdot \Pr[X = x] = \sum_{x>0} \Pr[X = x] = \Pr[X > 0].$$

- However,

$$\begin{aligned} \mathbb{E}[XY] &= \mathbb{E}\left[\sum_{i=1}^n X_i Y\right] = \sum_{i=1}^n \mathbb{E}[X_i Y] \\ &= \sum_{i=1}^n (\mathbb{E}[X_i Y \mid X_i = 1] \Pr[X_i = 1] + \mathbb{E}[X_i Y \mid X_i = 0] \Pr[X_i = 0]) \\ &= \sum_{i=1}^n \mathbb{E}[Y \mid X_i = 1] \Pr[X_i = 1] = \sum_{i=1}^n \mathbb{E}[1/X \mid X_i = 1] \Pr[X_i = 1] \\ &\geq \sum_{i=1}^n \frac{\Pr[X_i = 1]}{\mathbb{E}[X \mid X_i = 1]}. \quad (\text{by Jensen's inequality; } \mathbb{E}[f(Z)] \geq f(\mathbb{E}[Z])) \end{aligned}$$



An alternative proof of the 4-clique existence (1/3)

- Let $X = \sum_{i=1}^{\binom{n}{4}} X_i$, where X_i is 1 if the subset of four vertices C_i is a 4-clique and 0 otherwise.
- For a specific X_j , we have $\Pr[X_j = 1] = p^6$.



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- Let $X = \sum_{i=1}^{\binom{n}{4}} X_i$, where X_i is 1 if the subset of four vertices C_i is a 4-clique and 0 otherwise.
- For a specific X_j , we have $\Pr[X_j = 1] = p^6$. Using the linearity of expectations, we compute

$$\mathbb{E}[X \mid X_j = 1] = \mathbb{E} \left[\sum_{i=1}^{\binom{n}{4}} X_i \mid X_j = 1 \right] = \sum_{i=1}^{\binom{n}{4}} \mathbb{E}[X_i \mid X_j = 1].$$



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- Conditioning on $X_j = 1$, we now compute $\mathbb{E}[X_i \mid X_j = 1]$ by using that, for a 0-1 random variable,

$$\mathbb{E}[X_i \mid X_j = 1] = \Pr[X_i = 1 \mid X_j = 1].$$



An alternative proof of the 4-clique existence (2/3)

- There are $\binom{n-4}{4}$ sets of vertices C_i that do not intersect C_j . Each corresponding X_i is 1 with probability p^6 .



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- Similarly, $X_i = 1$ with probability p^6 for the $4\binom{n-4}{3}$ sets C_i that have one vertex in common with C_j .



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- Similarly, $X_i = 1$ with probability p^6 for the $4\binom{n-4}{3}$ sets C_i that have one vertex in common with C_j .
- For the remaining cases, we have
 - $\Pr[X_i = 1 \mid X_j = 1] = p^5$ for the $6\binom{n-4}{2}$ sets C_i that have two vertices in common with C_j and
 - $\Pr[X_i = 1 \mid X_j = 1] = p^3$ for the $4\binom{n-4}{1}$ sets C_i that have three vertices in common with C_j .



An alternative proof of the 4-clique existence (3/3)

- Overall, we have

$$\begin{aligned}\mathbb{E}[X \mid X_j = 1] &= \sum_{i=1}^{\binom{n}{4}} \mathbb{E}[X_i \mid X_j = 1] \\ &= 1 + \binom{n-4}{4} p^6 + 4 \binom{n-4}{3} p^6 + 6 \binom{n-4}{2} p^5 \\ &\quad + 4 \binom{n-4}{1} p^3.\end{aligned}$$

- Applying Theorem 8 yields (note: $\frac{\Pr[X_j=1]}{\mathbb{E}[X \mid X_j=1]}$ is the same $\forall j$)

$$\Pr[X > 0] \geq \frac{\binom{n}{4} p^6}{1 + \binom{n-4}{4} p^6 + 4 \binom{n-4}{3} p^6 + 6 \binom{n-4}{2} p^5 + 4 \binom{n-4}{1} p^3},$$

which approaches 1 as $n \uparrow$ when $p = f(n) = \omega(n^{-2/3})$.



Discussions

